DOMI-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 4: Double Trouble: Subitizing Doubles

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why does tile 6|6 fill the 12-frame tray completely?'
- Ask: 'How does knowing 4 + 4 = 8 help you solve 4 + 5?'
- Ask: 'Why do domino pips make patterns of 5 look like an X and 6 look like parallel lines?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students counting pips one-by-one. Encourage instant recognition: 'I see 5 and 5, so it must be 10!'

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"A mystery double tile has 8 pips in total. What tile is it?"
""

OFFICIAL SOLUTION KEY:

Tile 4|4 (since 4 + 4 = 8).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/math-dominoes/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution
within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

DOMI-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 1: Double Trouble: Subitizing Doubles

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why are doubles facts (4+4, 5+5, 6+6) the fastest facts in mathematics? Because your brain loves mirror symmetry. In this lab, find all 6 doubles tiles.

STEP-BY-STEP APPARATUS SETUP:

1. Spread all 28 dominoes face up on the table.
2. Find all the 'Double Tiles' where both ends match: 1|1, 2|2, 3|3, 4|4, 5|5, 6|6.
3. Line them up in ascending order.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: 28 Double-Six Dominos + 12-Frame Regrouping Tray

DATUM TARGET: Doubles Invariant: 2 × N is Always Even, Instantly Recognizable by Parallel Symmetry

CALCULATOR KEY: [3] [ENTER] [2] [*] -> 6 (Double 3)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 1A, Chapter 4 (Doubles & Addition)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 1, Unit 2 (Doubles & Domino Arrays)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 1, Ch. 2 • CCSS.MATH.1.OA.C.6 • TEKS 1.3D
GRADE LEVEL BAND:	Grades K-2 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 4 ('Operations & Skip-Counting'), pp. 72-80
HARDWARE UNBOXING BOOKLET:	DOMI-01 Booklet: Page 2 (Combinatoric Distribution) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

DOMI-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE SYMMETRY RACE

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Point to 1|1: 1 + 1 = 2.

Step 2: Point to 2|2: 2 + 2 = 4. Point to 3|3: 3 + 3 = 6.

Step 3: Point to 4|4: 4 + 4 = 8. Point to 5|5: 5 + 5 = 10. Point to 6|6: 6 + 6 = 12.

Step 4: Notice: Every single double total is an EVEN number (2, 4, 6, 8, 10, 12).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Doubles Invariant: 2 × N is Always Even, Instantly Recognizable by Parallel Symmetry

PHYSICAL APPARATUS MANIFEST:

28 Double-Six Dominos + 12-Frame Regrouping Tray

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

DOMI-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: BLIND GRAB DOUBLES DRILL

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: BLIND GRAB DOUBLES DRILL

THE NON-ROUTINE PUZZLE:

Partner A flashes a double tile for 1 second. Partner B calls out the sum.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Flash tile [4|4]. What is the sum?**
   Expected Physical Finding: 8

2. **Flash tile [6|6]. What is the sum?**
   Expected Physical Finding: 12 (Fills the 12-frame tray exactly!)

3. **Can an odd number ever be a double fact?**
   Expected Physical Finding: Never! A double is always two equal halves (even).

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [3] [ENTER] [2] [*] -> 6 (Double 3)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
3	X: 3	Base operand 3
ENTER	Y: 3 | X: 3	Stack lift
2	Y: 3 | X: 2	Scale factor 2
*	X: 6	Multiply by 2 evaluates double

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

DOMI-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 4: Near-Doubles & Compensation

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is it faster to do 6 + 6 + 1 than counting 7 dots after 6?'
- Ask: 'Are near-doubles sums always odd or always even?'
- Ask: 'How does this trick turn 12 hard math facts into instant mental recall?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students adding 1 to the wrong number. Emphasize: 'Double the smaller number and add 1, OR double the larger a
nd subtract 1.'

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Use near-doubles to solve 7 + 8 in your head. What is your thought process?"
""

OFFICIAL SOLUTION KEY:

Double 7 is 14. 14 + 1 = 15 (or double 8 is 16; 16 − 1 = 15).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/math-dominoes/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

DOMI-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 1: Near-Doubles & Compensation

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

What do you do if you forget what 5 + 6 is? If you know 5 + 5 = 10, then 5 + 6 is just 'one more'!

STEP-BY-STEP APPARATUS SETUP:

1. Pull tile 3|3 and tile 3|4 from your domino tray.
2. Place them side-by-side.
3. Compare their pip arrays.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Dominoes: Doubles 3|3, 4|4, 5|5 and Near-Doubles 3|4, 4|5, 5|6
DATUM TARGET:	Compensation Rule: N + (N + 1) = 2N + 1 (Doubles Plus One)
CALCULATOR KEY:	[4] [ENTER] [2] [*] [1] [+] -> 9 (4 + 5 = 2×4 + 1)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 1A, Chapter 4, Lesson 4.3 (Doubles Plus One)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 1, Unit 2, Module 2 (Near-Doubles Strategy)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 1, Ch. 2, Sec. 2.4 • CCSS.MATH.1.OA.C.6 • TEKS 1.3D
GRADE LEVEL BAND:	Grades 1–2 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 4, pp. 80–86
HARDWARE UNBOXING BOOKLET:	DOMI-01 Booklet: Page 5 (Tutorial 1) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

DOMI-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE PLUS-ONE EXTRA PIP

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Look at 3|3: exactly 3 on the left, 3 on the right = 6.

Step 2: Look at 3|4: it has 3 on the left, but 4 on the right. 4 is just 3 + 1!

Step 3: So 3|4 is just double 3 plus 1: (3 + 3) + 1 = 6 + 1 = 7.

Step 4: Now test tile 4|5: Double 4 is 8, plus 1 is 9! (4 + 5 = 9).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

- ✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
- ✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
- ✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
- ✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:

Compensation Rule: N + (N + 1) = 2N + 1 (Doubles Plus One)

PHYSICAL APPARATUS MANIFEST:

Dominoes: Doubles 3|3, 4|4, 5|5 and Near-Doubles 3|4, 4|5, 5|6

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

DOMI-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: DOUBLES MINUS ONE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: DOUBLES MINUS ONE

THE NON-ROUTINE PUZZLE:

Can you also solve 4 + 5 using Doubles MINUS One? (5 + 5 − 1).

FORM A MASTER STATION TASKS (Complete All 3):

1. **What is 5 + 5?**
   Expected Physical Finding: **10**
2. **Subtract 1: 10 − 1 = ?**
   Expected Physical Finding: **9**
3. **Does Doubles Plus 1 (8 + 1) match Doubles Minus 1 (10 − 1)?**
   Expected Physical Finding: **Both equal 9! Two paths to the same truth.**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [4] [ENTER] [2] [*] [1] [+] –> 9 (4 + 5 = 2×4 + 1)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
4	X: 4	Base double 4
2	Y: 4 | X: 2	Double factor
*	X: 8	Double total = 8
1	Y: 8 | X: 1	Plus one compensation
+	X: 9	Final near-double result = 9

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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DOMI-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 4: The 12-Frame Associative Train

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why are there so many ways to make 12?'
- Ask: 'How does the associative property let you rearrange the tiles in any order?'
- Ask: 'What happens if you try to make 12 using three identical tiles?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students counting blank ends as 1. Remind them: blank means ZERO pips!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Can you make 12 using three dominoes where every domino is a double tile?"
""

OFFICIAL SOLUTION KEY:

Yes: [1|1] (2) + [2|2] (4) + [3|3] (6) = 2 + 4 + 6 = 12!

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

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---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

DOMI-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 1: The 12-Frame Associative Train

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does the domino tray have exactly 12 sockets? Because 12 is the ultimate composite number, divisible by 1, 2, 3, 4, 6, and 12.

STEP-BY-STEP APPARATUS SETUP:

1. Dock the 12-Frame Regrouping Tray in the center of the table.
2. Notice: Tile [6|6] has exactly 12 pips and fills the entire tray capacity.
3. Challenge: Can you find TWO different dominoes that add up to exactly 12 pips?

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	12-Frame Tray + Dominoes [6|6], [5|5], [4|6], [3|6], [2|5], etc.
DATUM TARGET:	Associative Invariant: (A + B) + C = A + (B + C) = 12 (Full Tray Capacity)
CALCULATOR KEY:	[4] [ENTER] [5] [+] [3] [+] -> 12

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 1B, Chapter 10 (Addition to 20)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 2, Unit 1, Module 2 (Combinations to 12)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 1, Ch. 3 • CCSS.MATH.1.OA.B.3 • TEKS 1.3D
GRADE LEVEL BAND:	Grades 1-3 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 4, pp. 86-92
HARDWARE UNBOXING BOOKLET:	DOMI-01 Booklet: Page 2 (12-Frame Tray Geometry) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

DOMI-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: DOCKING DOMINO PAIRS

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Pull tile [4|3] (7 pips). What second tile is needed to make 12? (12 − 7 = 5 pips).

Step 2: Find a tile with 5 pips (e.g. [2|3] or [1|4] or [0|5]). Dock them both in the tray.

Step 3: Count total pips: 7 + 5 = 12! The pair balances the tray.

Step 4: Now find THREE dominoes that add to 12: e.g. [1|2] (3) + [2|2] (4) + [1|4] (5) = 12.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Associative Invariant: (A + B) + C = A + (B + C) = 12 (Full Tray Capacity)

PHYSICAL APPARATUS MANIFEST:

12-Frame Tray + Dominoes [6|6], [5|5], [4|6], [3|6], [2|5], etc.

💡 LABORATORY PRO-TIP:

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

DOMI-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: ALL-DOMINO 12-TRAIN

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: ALL-DOMINO 12-TRAIN

THE NON-ROUTINE PUZZLE:

How many unique pairs of dominoes in the box total 12 pips?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Pair 1: Tile [6|5] (11 pips) + ?**
   Expected Physical Finding: **Tile [0|1] or [1|0] (1 pip)**

2. **Pair 2: Tile [5|5] (10 pips) + ?**
   Expected Physical Finding: **Tile [1|1] or [0|2] (2 pips)**

3. **Pair 3: Tile [4|5] (9 pips) + ?**
   Expected Physical Finding: **Tile [0|3] or [1|2] (3 pips)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [4] [ENTER] [5] [+] [3] [+] -> 12

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
4	X: 4	First tile pip value
5	Y: 4 | X: 5	Second pip value
+	X: 9	Subtotal = 9
3	Y: 9 | X: 3	Third pip value
+	X: 12	Fills the 12-frame tray exactly

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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DOMI-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 4: The 28-Tile Triangular Pyramid

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why did row lengths shrink by exactly 1 in each step?'
- Ask: 'Why didn\'t we include tile [3|1] in the 3s row?' (Already included as [1|3]!).
- Ask: 'How does the pyramid prove that no dominoes were duplicated or forgotten?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to count [1|2] and [2|1] as two separate tiles. Explain that dominoes are unordered pairs—rota
ting the tile doesn't make a new piece!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"What is the total sum of all pips across the entire 28-tile set?"
""

OFFICIAL SOLUTION KEY:

Every number 0–6 appears 8 times. Total = 8 × (0+1+2+3+4+5+6) = 8 × 21 = 168 pips.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/math-dominoes/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution
within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

DOMI-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 1: The 28-Tile Triangular Pyramid

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does a domino set have exactly 28 tiles instead of 25 or 30? Because 28 is a triangular number formed by all possible pair combinations of numbers 0 through

STEP-BY-STEP APPARATUS SETUP:

1. Clear your table.
2. Your goal: Build the grand 7-row triangular domino pyramid.
3. Row 0 (Blanks): 7 tiles with a blank end: [0|0], [0|1], [0|2], [0|3], [0|4], [0|5], [0|6].

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Complete Set of 28 Double-Six Dominos + Table Workspace

DATUM TARGET: Combinatoric Invariant: Pairs from {0,1,2,3,4,5,6} with replacement = 7×8/2 = 28 Tiles

CALCULATOR KEY: [7] [ENTER] [8] [*] [2] [/] -> 28 (T_7 formula n(n+1)/2)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 3B, Chapter 14 (Patterns & Combinations)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 4, Unit 6 (Patterns, Arrays & Combinatorics)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 4, Ch. 10 • CCSS.MATH.4.OA.C.5 • TEKS 4.5B
GRADE LEVEL BAND:	Grades 3–5 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 7 ('Data & Probability'), pp. 162–170
HARDWARE UNBOXING BOOKLET:	DOMI-01 Booklet: Page 2 (T_7 Triangular Distribution) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

DOMI-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 2: LAB INVESTIGATION: BUILDING THE TRIANGLE

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Lay the 7 blank tiles in a long horizontal row at the bottom (Row 1: 7 tiles).

Step 2: Above it, lay all tiles starting with 1 (excluding [1|0] already used): [1|1], [1|2], [1|3], [1|4], [1|5], [1|6] (Row 2: 6 tiles).

Step 3: Row 3: All remaining tiles starting with 2: [2|2], [2|3], [2|4], [2|5], [2|6] (5 tiles).

Step 4: Continue upward: 4 tiles, 3 tiles, 2 tiles, and peak tile [6|6] (1 tile). Notice: 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28 tiles!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

- ✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
- ✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
- ✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
- ✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Combinatoric Invariant: Pairs from {0,1,2,3,4,5,6} with replacement = 7×8/2 = 28 Tiles

PHYSICAL APPARATUS MANIFEST:

Complete Set of 28 Double-Six Dominos + Table Workspace

💡 LABORATORY PRO-TIP:

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

DOMI-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: THE COMBINATORIC FORMULA

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE COMBINATORIC FORMULA

THE NON-ROUTINE PUZZLE:

Use the triangular formula T_n = n(n+1)/2 on your calculator.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Calculate T_7: 7 × 8 / 2.**
   Expected Physical Finding: **28 tiles (Double-Six Set)**
2. **How many tiles would a Double-Nine domino set have? (n = 10 elements: 0 to 9).**
   Expected Physical Finding: **10 × 11 / 2 = 55 tiles**
3. **How many tiles in a Double-Twelve set? (n = 13).**
   Expected Physical Finding: **13 × 14 / 2 = 91 tiles**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [7] [ENTER] [8] [*] [2] [/] -> 28 (T_7 formula n(n+1)/2)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
7	X: 7	Elements {0,1,2,3,4,5,6}
8	Y: 7 | X: 8	n + 1
*	X: 56	Permutation product
2	Y: 56 | X: 2	Symmetry divisor
/	X: 28	7th Triangular Number T_7 = 28 tiles

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

DOMI-01 • TENT 5/6 | ADVANCED | 15 Min

PANEL 4: Turnaround Facts & Commutativity

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why does rotation prove addition is commutative but not subtraction?'
- Ask: 'How does knowing turnaround facts cut your memorization list in half?'
- Ask: 'Does multiplication work like addition when you rotate a tile array?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students thinking 2 − 5 is 3. Point out that having 2 apples and eating 5 is impossible in real physical space without owing 3.

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"A student knows 3 + 8 = 11. What other addition fact do they know for free?"
""

OFFICIAL SOLUTION KEY:

8 + 3 = 11 (and fact family subtractions 11 − 3 = 8 and 11 − 8 = 3).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/math-dominoes/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
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- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

DOMI-01 • TENT 5/6 | ADVANCED | 15 Min

PANEL 1: Turnaround Facts & Commutativity

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does 2 + 5 equal 5 + 2? You don't need a math degree to prove it—just turn the domino upside down in your hand.

STEP-BY-STEP APPARATUS SETUP:

1. Pick up tile [2|5] with your left hand.
2. Look at it: 2 pips on the left, 5 pips on the right (2 + 5 = 7).
3. Rotate the tile 180° in your fingers.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Dominoes [2|5], [3|4], [1|6], [2|4] + Stacking Desk

DATUM TARGET: Geometric Invariant: 180° Spatial Inversion Preserves Total Pip Mass (A + B = B + A)

CALCULATOR KEY: [2] [ENTER] [5] [+] [5] [ENTER] [2] [+] [−] -> 0

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 1A, Chapter 3 (Commutative Property of Addition)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 1, Unit 1, Module 4 (Turnaround Facts)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 1, Ch. 3, Sec. 3.2 • CCSS.MATH.1.OA.B.3 • TEKS 1.3E
GRADE LEVEL BAND:	Grades 1–3 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 4, pp. 92–98
HARDWARE UNBOXING BOOKLET:	DOMI-01 Booklet: Page 6 (Aha! Discovery: Rotational Commutativity)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

DOMI-01 • TENT 5/6 | ADVANCED | 15 Min

PANEL 2: LAB INVESTIGATION: THE 180° ROTATION PROOF

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Look at the rotated tile: now 5 pips are on the left, and 2 are on the right (5 + 2 = 7).

Step 2: Did any pips appear out of thin air? Did any pips vanish? (No!).

Step 3: Conclude: The physical object is 100% identical. Therefore, 2 + 5 MUST equal 5 + 2.

Step 4: This is the Commutative Property of Addition (Turnaround Fact).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Geometric Invariant: 180° Spatial Inversion Preserves Total Pip Mass (A + B = B + A)

PHYSICAL APPARATUS MANIFEST:

Dominoes [2|5], [3|4], [1|6], [2|4] + Stacking Desk

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

DOMI-01 • TENT 5/6 | ADVANCED | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: DOES COMMUTATIVITY WORK FOR SUBTRACTION?

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: DOES COMMUTATIVITY WORK FOR SUBTRACTION?

THE NON-ROUTINE PUZZLE:

Test 5 − 2 vs 2 − 5 using the same domino.

FORM A MASTER STATION TASKS (Complete All 3):

1 **Calculate 5 − 2.**
Expected Physical Finding: 3

2 **Try to take 5 pips away from 2 pips.**
Expected Physical Finding: Cannot do it with physical pips! (Yields −3, not 3).

3 **Is subtraction commutative?**
Expected Physical Finding: NO! Order matters for subtraction.

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [2] [ENTER] [5] [+] [5] [ENTER] [2] [+] [−] –> 0

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
2	X: 2	Operand A
5	Y: 2 | X: 5	Operand B
+	X: 7	Forward sum = 7
5	Y: 7 | X: 5	Reverse operand B
2	Z: 7 | Y: 5 | X: 2	Reverse operand A
−	[illegible]	[illegible]

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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DOMI-01 • TENT 6/6 | ADVANCED | 20 Min

PANEL 4: The Difference Race: Subtractive Fluency

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why do all double tiles have a difference of zero?'
- Ask: 'Which tile gives the highest possible difference score?' ([6|0] = 6 points).
- Ask: 'How does finding the difference between two ends train your brain for division remainders?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students adding instead of subtracting. Emphasize: 'Difference means: how many MORE pips are on the big side t
han the small side?'

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Two dominoes are drawn: Tile A is [5|1], Tile B is [6|3]. Which tile has the greater difference?"
""

OFFICIAL SOLUTION KEY:

Tile A difference = 5 − 1 = 4. Tile B difference = 6 − 3 = 3. Tile A wins!

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

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at the QR code to launch the interactive 3D model,
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https://stackcalc.io/products/math-dominoes/#product-videos

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- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
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- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

DOMI-01 • TENT 6/6 | ADVANCED | 20 Min

PANEL 1: The Difference Race: Subtractive Fluency

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Who has the fastest mental subtraction in the class? In this game, flip dominoes and subtract the smaller side from the larger side before your partner.

STEP-BY-STEP APPARATUS SETUP:

1. Shuffle all 28 dominoes face down in a draw pile ('The Boneyard').
2. Each player draws 5 dominoes and keeps them hidden.
3. Rule: The score of any domino is the DIFFERENCE between its two ends: High − Low.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Complete 28-Tile Domino Set + Timer / SC-6 Calculator

DATUM TARGET: Difference Invariant: Difference = High Pip Count − Low Pip Count (Non-Negative Range 0–6)

CALCULATOR KEY: [6] [ENTER] [2] [−] -> 4 (Tile 6|2 Difference)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 2A, Chapter 2 (Mental Subtraction Strategies)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 2, Unit 2 (Difference as Distance on a Number Line)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 2, Ch. 4 • CCSS.MATH.2.OA.B.2 • TEKS 2.4A
GRADE LEVEL BAND:	Grades 1–4 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 4, pp. 98–104
HARDWARE UNBOXING BOOKLET:	DOMI-01 Booklet: Page 7 (RPN Postfix Connection) & Page 8 (Exercises)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

DOMI-01 • TENT 6/6 | ADVANCED | 20 Min

PANEL 2: LAB INVESTIGATION: SCORING DOMINO DIFFERENCES

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Inspect tile [6|1]: 6 − 1 = Difference of 5 points.

Step 2: Inspect tile [4|4]: 4 − 4 = Difference of 0 points (Double tiles score zero!).

Step 3: Inspect tile [5|2]: 5 − 2 = Difference of 3 points.

Step 4: Notice: The maximum possible difference in a Double-Six set is 6 (Tile [6|0]).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Difference Invariant: Difference = High Pip Count − Low Pip Count (Non-Negative Range 0–6)

PHYSICAL APPARATUS MANIFEST:

Complete 28-Tile Domino Set + Timer / SC-6 Calculator

💡 LABORATORY PRO-TIP:

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

DOMI-01 • TENT 6/6 | ADVANCED | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: THE TARGET DIFFERENCE RACE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE TARGET DIFFERENCE RACE

THE NON-ROUTINE PUZZLE:

Find all dominoes in the set that have a difference of exactly 2.

FORM A MASTER STATION TASKS (Complete All 3):

1. **List all difference-2 tiles.**
   Expected Physical Finding: [2|0], [3|1], [4|2], [5|3], [6|4]
2. **How many difference-2 tiles exist?**
   Expected Physical Finding: Exactly 5 tiles
3. **How many difference-0 tiles (doubles) exist?**
   Expected Physical Finding: Exactly 7 tiles ([0|0] through [6|6])

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [6] [ENTER] [2] [−] -> 4 (Tile 6|2 Difference)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
6	X: 6	High pip side
ENTER	Y: 6 | X: 6	Stack lift
2	Y: 6 | X: 2	Low pip side
−	X: 4	Subtract evaluates difference between ends

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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